

Proposal: Modular DC Fan Array for Airflow Management in AWG Systems

Objective:

To implement a modular DC fan array system to improve airflow management in Atmospheric Water Generators (AWG) systems. This system aims to reduce power consumption, enhance airflow efficiency, and simplify maintenance. Very high CFM and static pressure output for significantly lower power consumption. Greatly increasing scalability whilst reducing the impact of this tech.

2. Background

Current Potential Challenges:

- High power consumption of single large industrial motors.
- Complexity and difficulty in maintaining large motors.
- Inefficiency in airflow management, leading to increased operational costs.

Proposed Solution:

Utilizing a modular DC fan array with a plenum to improve airflow distribution, reduce power consumption, and facilitate easier maintenance. Allows for easy calculation of system designs for larger applications. The reasoning behind this is due to fan law 3 showing that power consumption is the cube of fan speed – the faster a fan moves, the more power it requires to drive depreciating CFM and static pressures.

Limitations:

Has strategic limitations in smaller applications. The motors are lower torque axial fans and thus require the use of plenums versus ducting to efficiently deliver air throughout the absorption media. Calculating ideal fan spacing for the fans in parallel to not compete for air, using infeed ducting, and output ducting to take the rotational speed of the air and drive it forward. Stator Vanes can be utilized, or counter rotating fans in parallel, which would result in both, high cfm with greatly increased kinetic energy of the air post exhaust and higher static pressure.

Example Setup:

Nidec GentleTyphoon fans, with 6 fans in series and 24 fans in parallel within an ideal system, provides a substantial airflow of 1875 CFM and an effective static pressure of 22.05 mmH₂O, while maintaining a low total power consumption of approximately 29.88 W. This design ensures efficient operation, enhanced performance, and ease of maintenance, making it an ideal solution for the specified application in AWG systems.

3. Technical Explanation

Fan Laws:

Fan Law 1

$$CFM_2 = CFM_1 \times \left(\frac{RPM_2}{RPM_1} \right)$$

Fan Law 2

$$SP_2 = SP_1 \times \left(\frac{RPM_2}{RPM_1} \right)^2$$

Fan Law 3

$$HP_2 = HP_1 \times \left(\frac{RPM_2}{RPM_1} \right)^3$$

-Fan Law 1:

Airflow (CFM) is directly proportional to fan speed (RPM). Using multiple fans at lower RPMs increases total CFM without the inefficiencies of a single high-speed fan.

-Fan Law 2:

Static pressure is proportional to the square of the fan speed. Increasing the number of fans increases the static pressure more efficiently than increasing the speed of one fan.

- Fan Law 3:

Power consumption is proportional to the cube of the fan speed. Reducing fan speed significantly decreases power consumption.

4. System Design

Reliability and Redundancy:

- The system is designed to maintain functionality even in the event of individual fan failures, ensuring operational continuity with reduced volume or static pressure.
- A critical spares inventory, including rev-locked, pre-programmed PLCs, allows for quick replacement and system reboot if a PLC fails.
- Modular DC power modules can be swapped easily, with accessible bit terminals, ensuring rapid recovery from any input or output failure.

Simplified Maintenance and Support:

- The modular system design facilitates easy maintenance and rapid troubleshooting, reducing complexity and minimizing potential errors.
- Flow diagrams guide maintenance operations, helping technicians efficiently troubleshoot and repair.

Components:

1. DC Fan Array:

- Multiple 120mm DC fans, such as Nidec GentleTyphoon, arranged in a modular setup.
- Arrays of fans in parallel to maximize CFM and in series to increase static pressure as needed.

2. Plenum:

- Designed to reduce resistance and distribute air evenly across the absorption medium.

3. Wiring and Monitoring:

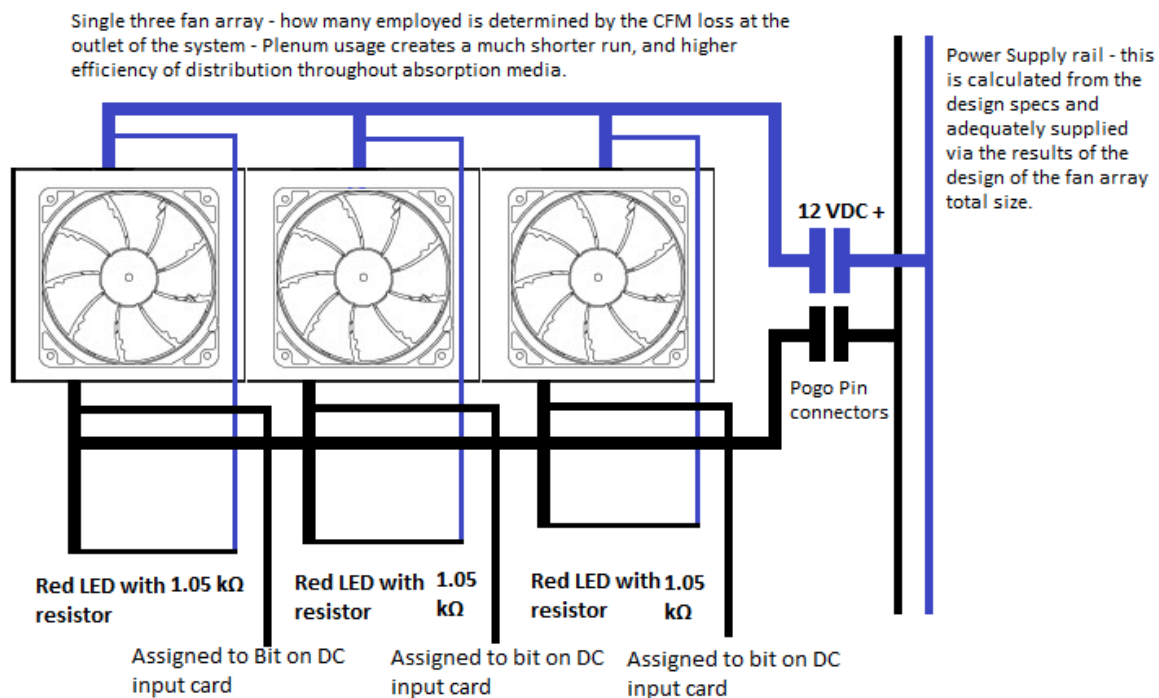
- Parallel wiring for reliability.
- Monitoring system using PLC to detect and manage fan faults.

4. Array Housings – Plastic Injection Molded housing with a tongue and groove expandable system, internally houses the wires and the fans would be connecting through the internal chassis via Pogo Pin systems on the chassis out to the supply wiring for the Parallel System. Can be paired with a termination connection where necessary. Series wiring housings would be fans feeding into each other so linear assembly with each fan on a quick connect chassis to hot swap in and out as needed.

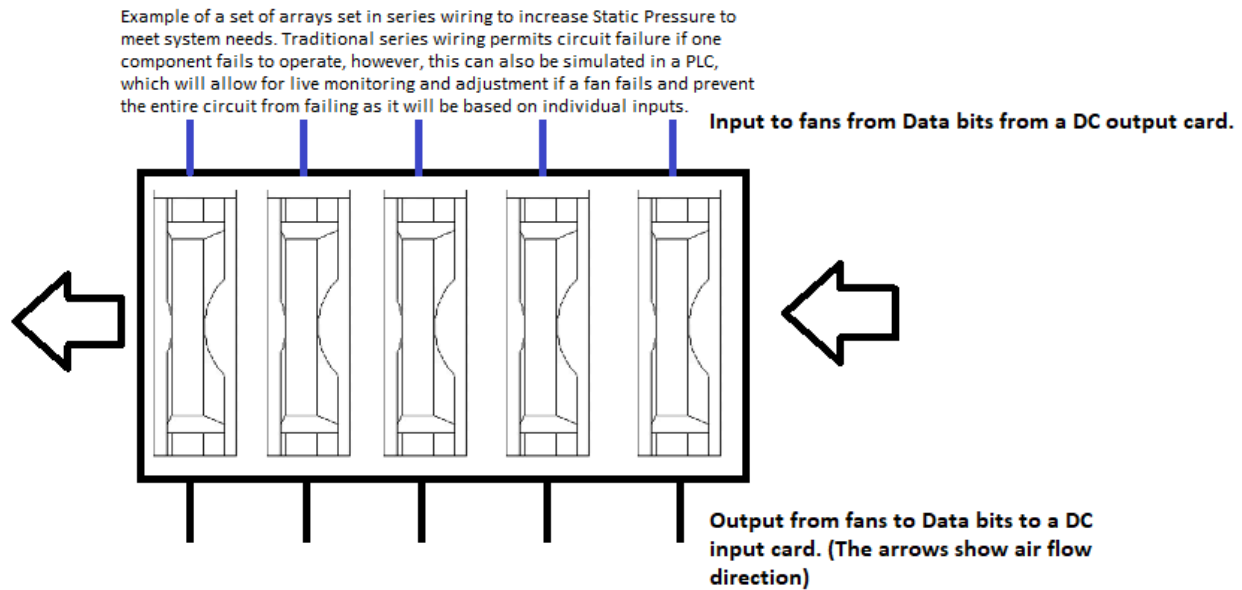
Wiring Diagrams:

- Parallel Fan Array Wiring

For simplicity in display this was calculated on fans running on 12 volts. The resistor rating is going to change for the specific voltages employed.



- Series Fan Array Wiring (simulated via PLC)



5. Ladder Logic for PLC Control

- <https://app.plcsimulator.online/puyUxSNsyOyBKTO4yRyu>
- System is designed to operate on a “total voltage divided by available fans” to output the appropriate voltage per fan. As/if a fan fails the system automatically calculates the new target voltage for each remaining fan. The status of the Array fans is included however these are voltage controlled directly – unless integration via a PWM (pulse width module) was desired for more precise synchronization – however this is unnecessary if these are wired up correctly – they will all carry the same voltage. Voltage is calculated by a total amps based on the total arrays in the system.
- Manometers and Anemometers may be introduced throughout the system for live monitoring on the HMI and feedback into the PLC for live voltage adjustments. These will also help determine from the size of the exchanger and the resistance of the absorption material the amount of fans to be employed in system design allowing for a modular, or stackable expandability based on required fluid output.

6. Benefits

Expanded System Capabilities:

- Environmental filters, such as expandable NBC filters, enhance adaptability for various applications, increasing market appeal.
- The system integrates seamlessly with alternative energy sources like solar and wind, using direct DC power without inefficient rectifiers.

Reduced Power Consumption:

- Lower power requirements by using multiple fans at lower RPMs.

Enhanced Efficiency:

- Improved airflow distribution and reduced resistance via a well-designed plenum.

Simplified Maintenance:

- Easy hot-swapping of fan arrays for quick repairs and maintenance.

Scalability:

- Modular design allows for easy scalability based on system requirements.

7. Strategic Vision: Net Zero Architecture Integration

Net Zero Integration:

- Modular DC fan arrays can be integrated into net-zero architecture, supporting sustainable and energy-efficient building designs.
- Current issues with high efficiency cooling within the Z6 Tower – creates significant issues with condensation when in humid areas. There is a need to dehumidify the air – This creates a dynamic niche for this product to be integrated readily, into current net zero construction efforts.

Market Trends:

- With the AI boom and increased demand for data centers, the need for efficient cooling and water generation systems is critical.

National Strategic Interests:

- Aligning with global leaders like China in net-zero infrastructure to enhance national capabilities in sustainable development.

8. Conclusion

Summary:

- The proposed modular DC fan array system offers significant improvements in efficiency, power consumption, and maintenance over traditional single-fan systems. By leveraging fan laws and modern PLC monitoring, this system is poised to enhance the functionality and reliability of AWG systems.